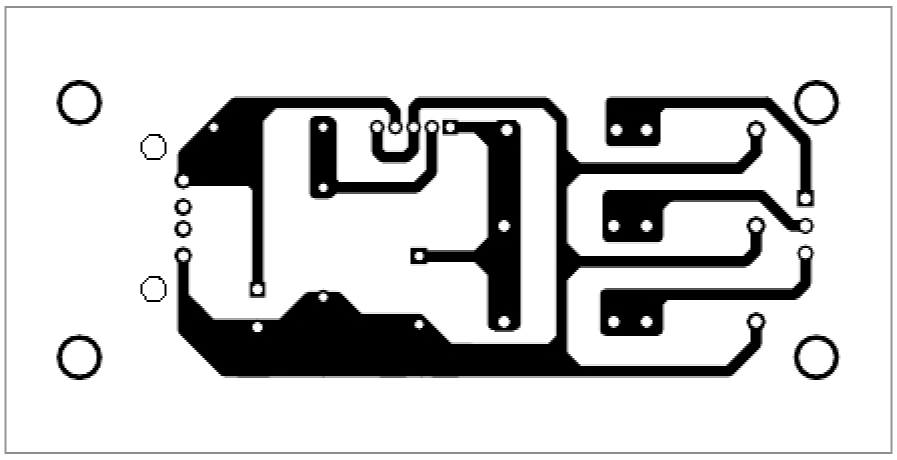

Fig. 4: Actual-size PCB layout for hydro power generation

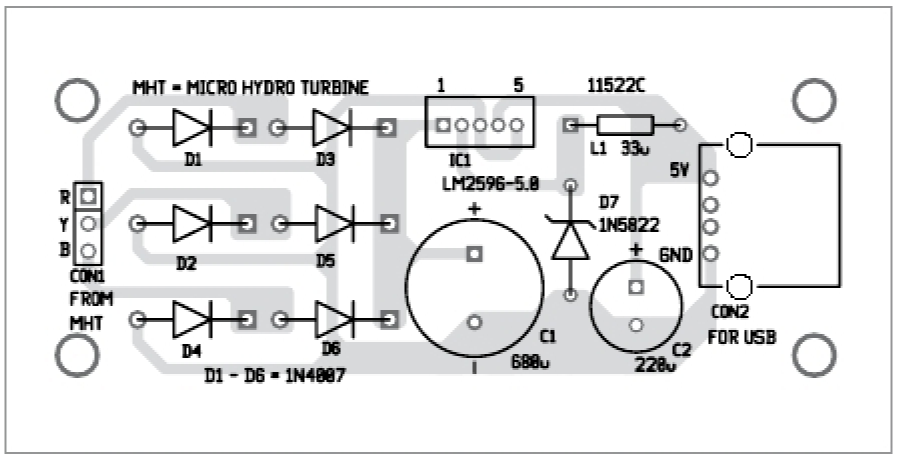

Fig. 5: Components layout for the PCB

LM2596 series offers a high-efficiency replacement for the popular three-terminal linear regulators. It substantially reduces the size of the heat-sink, and in some cases no heat-sink is required.

Requiring a minimum number of external components, this regulator is simple to use and includes fault protection and a fixed-frequency oscillator. LM2596 is the upgraded version of IC LM2576, which has 52kHz switching frequency. LM2596 has 150kHz switching frequency that offers usage of low-value capacitors and filters.

Micro hydro-turbine. A generic MHT generator is used in this project. It has maximum output voltage of about 20V, three-phase AC, maximum output current of about 150mA with flow rate of about 2.5 to 25 litres per minute.

Construction and testing

An actual-size PCB layout for hydro power generation from a water pipe is shown in Fig. 4 and its components layout in Fig. 5. Assemble the circuit on the designed PCB. Connect MHT RYB outputs across CON1 using external wires. Power supply for the circuit is provided by MHT.

After assembling the circuit on the PCB, fix MHT with a suitable sized water pipe—size of the water pipe depends on the coupling heads of MHT. When water starts flowing through the pipe continuously, you should be able to get 5V DC across CON2. EFY

A. Samiuddhin is B.Tech in electrical and electronics engineering. His interests include LED lighting, power electronics, microcontrollers and Arduino programming